Why Join W3A Foundation Lux Network platform access and patent license through W3A Foundation membership

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Executive summary

W3A Foundation (`w3a.foundation`) is the licensing collective for the Lux Industries patent estate and the operational on-ramp to the Lux Network institutional substrate. Membership provides three durable benefits: (i) a license to the full Lux patent estate, including the foundational 2023 rollup-architecture patent and the 15-application post-quantum patent family filed March 3, 2026; (ii) production access to the Lux Network substrate purpose-built for regulated digital finance; and (iii) a participating seat in the standards-setting work that will define the next decade of institutional tokenization infrastructure.

No automatic licenses. No grandfathering. The patent estate covers every commercially relevant surface in regulated digital finance — Layer-1 consensus, rollups, cross-chain messaging, exchange / ATS matching, broker-dealer onboarding, transfer-agent cap tables, fiat on/off-ramps, fund administration, atomic T+0 settlement, custody, FHE-confidential compliance, and DEX architectures — and enforcement against non-members commences in 2026.

This paper outlines what W3A Foundation provides, the structure of membership, the confidentiality framework that allows substantive evaluation without IP exposure, and the process by which an institutional buyer becomes a member.

1 What W3A Foundation is

W3A Foundation is the standards body and patent-licensing collective for the Lux ecosystem. Three roles.

Licensing collective. The foundation administers the patent license that flows to members from Lux Industries Inc., the patent holder. Members receive a perpetual, field-of-use-restricted license to practice the entire patent estate under the foundation's standard membership terms.

Standards body. The foundation hosts the technical standards work that defines how regulated digital finance operates on the Lux substrate and on adjacent licensed infrastructure. Member organizations participate in the standards committees shaping ERC-3643 + OnchainID profiles, post-quantum migration profiles, atomic-settlement profiles, custody profiles, and FHE-confidential-compliance profiles.

Ecosystem coordinator. The foundation coordinates participating commercial tenants and licensed third-party operators, manages defensive cross-license inventory across the member pool, and runs the certification program that attests to members' regulatory and operational posture.

The foundation is structured as a non-profit Delaware entity, governed by a board representing founding members, with administrative functions performed under standard intercompany agreements. Members do not own or control Lux Industries, the patent holder; the patent estate is owned by Lux Industries Inc. and licensed to the foundation, which sublicenses to members.

2 What members receive

2.1 Patent license

Members receive a license to the full Lux patent estate within their selected field of use. The license is perpetual for the term of the underlying patents, royalty-free under the membership-fee arrangement, non-transferable, non-exclusive but field-restricted within the member's commercial domain, and includes:

- Practice rights for every patent in the estate (the 2023 foundational patent and the 15-application 2026 post-quantum family) within the selected field of use.
- Have-made rights covering members' appointed agents acting on the member's behalf (registered transfer agents, qualified custodians, named technology providers).
- Defensive cross-license-back covenant on patents the member develops within the relevant fields.
- No-challenge covenant for the term of membership, with defensive termination if the member asserts patents against Lux Industries or against any other foundation member.
- Most-favored-licensee protection within the member's selected field for Strategic and Founding tiers.
- Coverage of the post-quantum migration path through the 2026 family, providing operational certainty for the 2027–2032 regulatory PQ migration window.

2.2 Substrate access (optional)

Members may additionally elect substrate access — the ability to run their tokenized operations directly on Lux Network mainnet validators, with Lux KMS for HSM-backed secret management, Lux IAM for OAuth2 / OIDC, Lux Gateway for JWT-validated request routing, and Lux Operator Kubernetes CRD for declarative deployment. Substrate access is a separate fee tier negotiated alongside membership and is the preferred deployment path for members building tokenized funds, regulated trading platforms, or institutional custody offerings.

2.3 Standards participation

Members participate in W3A Foundation technical standards committees. Member representatives vote on standards profile updates, propose new standards work items, and receive early access to standards drafts before public publication. For institutional buyers building long-lived tokenization infrastructure, standards participation is durable competitive advantage — the member shapes the rules the rest of the industry will eventually follow.

2.4 Certification program

Members may apply for W3A-Certified status on specific operational deployments. Certification combines a technical audit of the deployment against W3A standards profiles with an attestation by the member's compliance officer to ongoing operational adherence. W3A-Certified deployments may use the W3A certification mark in customer-facing materials and benefit from the foundation's coordination with prudential regulators and external auditors.

2.5 Defensive coalition

Every member commits to cross-license-back to the foundation on patents the member develops. Over time the foundation accumulates a substantial defensive portfolio derived from members' own innovations. Member participation in the cross-license coalition protects the member against future IP assertion from any other member and from the foundation itself.

3 The Lux Network substrate

A member's commercial deployments on the Lux substrate access the following capabilities.

Quasar consensus. Post-quantum Layer-1 consensus with parallel BLS + Pulsar + Corona certificates. Configurable profiles: classical-only, parallel-classical-plus-PQ, hybrid, strict-PQ, maximum-strict. The substrate is post-quantum-ready today, with strict-PQ profile drop-in available for members operating under prudential PQ mandates.

Lux EVM. EVM-compatible execution layer with native precompiles for ML-DSA, SLH-DSA, FN-DSA, hybrid PQ+classical, FROST and CGGMP21 threshold signatures, BLS aggregate, Ed25519, secp256r1, FHE primitives (BFV / CKKS / TFHE), and the Lux atomic-settlement gateway. Members deploy standard EVM smart contracts with access to PQ verification primitives at gas costs amortized by the precompile architecture.

Native compliance. ERC-3643 + OnchainID identity registries, trusted-issuer registries, Resale-Quadrant / CrowdfundFirstYear / OffshoreDistributionPeriod compliance modules, and transfer-agent signing authority as a protocol-level primitive. Compliance is not bolt-on; it is the platform.

Threshold MPC custody. CGGMP21 ECDSA and FROST EdDSA threshold signing with transfer-agent share as cryptographic constituent (not procedural prerequisite). 2-of-3 user wallets and 3-of-5 treasury wallets as standard configurations, with cryptographically-enforced regulated custody.

FHE-confidential compliance. Encrypted-jurisdiction predicates, encrypted-accreditation status checks, and encrypted-sanctions screening through FHE precompiles. Members may run compliance gates that do not reveal counterparty identity information on-chain.

Atomic T+0 settlement. The Lux-native atomic mint-and-burn pair settlement primitive provides instantaneous T+0 settlement of tokenized securities trades with all-or-nothing atomicity. Operationally simpler than DTC + DTCC; cryptographically auditable; per-trade reversible only through transfer-agent-authorized reversal.

Cross-chain bridging. Native Lux Bridge for institutional flows; license-protected interoperability with major cross-chain messaging protocols. Members deploy on Lux substrate while retaining cross-chain availability to ecosystem counterparties.

Operational maturity. Lux Network has been in production with commercial tenants since 2024. Production controls (push-only ACH, name-matching, velocity caps, multi-processor failover, withdrawal hold windows) are battle-tested at production scale across regulated U.S. securities and spot crypto markets.

4 The IP estate behind the license

W3A Foundation membership grants license to the following.

Foundational acquired patent. U.S. Patent 11,803,853 B2, "Controlling transactions on a network," priority 4 May 2018, granted 31 October 2023, acquired by Lux Industries Inc. in 2025.

Foreign counterparts under diligence in EP, UK, Norway, Japan, Singapore, UAE, Switzerland. Expires 6 May 2039.

Post-quantum continuation family, filed March 3, 2026.

Application	**Subject matter**
2026-001 (CIP)	Master PQ extension of 11,803,853
2026-002	PQ data-availability attestation (L2 / L3 rollups)
2026-003	PQ cross-chain message gating
2026-004	PQ institutional tokenization settlement (general)
2026-005	Cryptographic algorithm-agility framework
2026-006	PQ threshold MPC for regulated custody
2026-007	PQ parallel-certificate L1 consensus
2026-008	PQ ATS / exchange matching and settlement
2026-009	PQ broker-dealer customer onboarding and funding
2026-010	PQ transfer-agent cap table and corporate actions
2026-011	PQ banking fiat on-ramp and off-ramp
2026-012	PQ tokenized fund administration and NAV
2026-013	PQ atomic T+0 securities settlement
2026-014	PQ + FHE confidential compliance gate
2026-015	PQ DEX / order-book and AMM with compliance gating

The estate covers L1 consensus, L2 rollups, cross-chain messaging, exchange / ATS, broker-dealer onboarding, transfer-agent cap-tables, fiat on/off-ramps, fund administration, atomic settlement, custody, FHE-confidential compliance, DEX architectures, and the algorithm-agility framework that ties them together. Coverage extends through 2046+ depending on prosecution timing.

4.1 Why the estate is durable

Priority date precedence. The foundational 4 May 2018 priority predates every modern rollup architecture in production. The 3 March 2026 PQ family priority predates the regulatorily-mandated migration window.

Plasma-and-MVP-aware claim construction. Lux's claim construction distinguishes over the strongest pre-2018 prior art (Plasma white paper, August 2017; Minimal Viable Plasma, January 2018) through threshold formalization, the completeness/correctness dichotomy, and the two-agent first-agent-plus-second-agent architecture. Continuation strategy provides backstop.

Primitive-agnostic and architecture-agnostic PQ family. The PQ family does not specify a single primitive. It claims the integration of post-quantum or hybrid signature primitives, plus FHE, into the broader transaction-control architecture. Every PQ migration path enters the patent zone regardless of which specific primitive is chosen.

Ten-year inventor track record. Lux Industries leadership has been building these primitives publicly since 2015. The `github.com/arcalabs` repository archive provides documented prior-art and written-description support for the patent claims, with the arc starting at the `arcadia` desktop wallet (August 2015) and continuing through `ethereum-cluster` (April 2018), `st-contracts` Arca Security Token protocol (January 2019), and `exchange-sdk` (November 2019). Ten-year inventor history establishes credibility against any derivation claim by later entrants.

5 Membership tiers

Tier	Audience	Annual fee	License scope
Founding	Strategic institutional partners; significant cross-license inventory and standards-leadership commitment	Negotiated	Full estate, perpetual, MFN, broad field of use
Strategic	Major institutional asset managers, regulated exchanges, large tokenization platforms	\$5M–\$25M / yr	Full estate, MFN within selected field of use
General	Regulated operators of ATS / BD / TA / fund admin / banking on-ramp / tokenized fund / DEX / RaaS	\$500K–\$5M / yr	Full estate within selected field of use
Associate	Early-stage operators, individual developers, academic and research participants	\$25K–\$250K / yr	Limited estate, narrow field of use

Field-of-use selection. Each member selects a field at membership time. Multiple fields are aggregated under a single Strategic or Founding membership; narrow single-field memberships qualify for General or Associate tier. Standard fields:

- Regulated ATS / exchange operations
- Broker-dealer customer operations
- Transfer-agent cap-table operations
- Banking fiat on/off-ramp
- Tokenized fund administration
- Atomic securities settlement
- FHE-confidential compliance
- DEX / order-book / AMM with compliance gating
- Cross-chain messaging operations
- L2 rollup and Alt-DA operations
- L1 consensus operations
- Threshold MPC custody operations
- Cryptographic-agility framework operations

Members may add fields by graduating to a higher tier at any time.

6 Confidentiality and NDA framework

The W3A Foundation engagement model is designed to allow substantive evaluation of the IP estate and the platform without IP exposure to the prospective member. The framework operates in four stages.

6.1 Stage 1 — Open conversation (no NDA)

Initial scoping conversations occur without confidentiality obligation. The foundation shares at this stage only publicly-available information: the existence of the foundational patent (public record), the existence of the W3A Foundation, the substrate technology positioning, the standards-program scope, and the membership-tier framework. The materials in this paper are at the Stage 1 disclosure level and may be shared with the prospective member's executive team without confidentiality obligation.

6.2 Stage 2 — Mutual NDA covering preliminary diligence

When both parties express material interest, a standard mutual non-disclosure agreement is executed covering the next phase of evaluation. Under the Stage 2 NDA, the foundation discloses to the prospective member:

- The complete inventory of the patent estate with priority dates, filing dates, and field-of-use mappings.
- Architecture-level descriptions of the Lux substrate including consensus, EVM precompiles, native compliance modules, and operational tooling.
- The membership term-sheet template with proposed fee structure, license scope, and cross-license-back terms for the prospective member's tier.
- The standards roadmap for the next 12–18 months including upcoming standards work items the prospective member would participate in.
- Anonymized prior-engagement information sufficient for the prospective member to evaluate the foundation's commercial track record.

Information protected under the Stage 2 NDA includes the foundation's confidential operational details, other members' specific terms, and Lux Industries' enforcement-program internal sequencing. The Stage 2 NDA is mutual: the prospective member's information shared with the foundation is equally protected.

6.3 Stage 3 — Tightened NDA covering full diligence

Following Stage 2 alignment on commercial structure, a tightened NDA with more specific information-protection provisions enables full technical and legal diligence:

- Full patent prosecution histories from the USPTO public file wrappers consolidated for the prospective member's counsel.
- Lux Industries' internal claim charts mapping each patent claim element to standard implementations.

- The prior-art defense memorandum addressing Plasma, Minimal Viable Plasma, Truebit, Chainlink, and other key references.
- The substrate technical reference covering deployment architectures, operational SLAs, and integration paths for the prospective member's existing infrastructure.
- The complete membership agreement template ready for outside-counsel review and countermark.

Information protected under the Stage 3 NDA includes Lux Industries' continuation-application content (pending applications not yet published), specific enforcement-program timing and targets, and the foundation's internal cross-license inventory.

6.4 Stage 4 — Executed membership agreement

Membership execution transitions the prospective member from NDA-protected evaluation to executed-license participation. Standard membership provisions include perpetual confidentiality of the prospective member's commercial deployment specifics, member-to-member confidentiality on competitive deployment information, and ongoing confidentiality of the foundation's internal operational data.

6.5 IP protection within the NDA framework

The NDA framework is designed so that the prospective member receives sufficient information at each stage to make an informed decision on the next stage, without receiving information that would enable the prospective member to design around the patent estate or to mount an effective invalidity challenge. Specifically:

- The patent claims themselves are public; the foundation does not protect what is already public.
- The foundation's internal claim charts and prior-art defense memoranda are protected; the prospective member receives these only at Stage 3 once material commercial alignment is in place.
- Continuation-application content covering future-filed claims is protected through Stage 4; even members do not receive complete continuation visibility, only the relevant portions for their field of use.

The framework is consistent with institutional norms for confidential IP licensing and has been calibrated to balance prospective-member evaluation needs against the foundation's legitimate confidentiality interests.

7 Engagement process

Stage	Duration	Output
1. Executive introduction	One conversation, 60–90 min	Open dialogue; this paper as reference
2. Mutual NDA + scoping diligence	2–4 weeks	Membership tier identified; field of use sc
3. Tightened NDA + full diligence	8–12 weeks	Detailed term sheet; outside counsel revie

4. Definitive membership agreement	4–6 weeks	Signing; public announcement at member
5. Substrate onboarding (if elected)	60–90 days	Production deployment of member's toke

Total time from initial conversation to executed membership for a Strategic-tier institutional buyer: typically 90 to 120 days. Accelerated timelines available for institutional buyers with executive sponsor and outside counsel in place at engagement.

8 Why now

Tokenization is institutional. BUIDL crossed $1.7B AUM in twelve months. Franklin Templeton BENJI, Ondo OUSG and USDY, Hamilton Lane SCOPE, KKR Health Care, Apollo Diversified Credit, and ten other named institutional tokenized vehicles are scaling now. Forrester, McKinsey, and BCG forecast $5–16T tokenized assets by 2030.

Post-quantum migration is regulatorily mandated. NIST FIPS 203/204/205 (2024) and HQC (2025). OCC, Federal Reserve, SEC, ECB, BSI, NCSC, MAS guidance for the financial-sector PQ migration window 2027–2032. Every institutional digital-finance platform will migrate. The W3A patent license covers every migration path.

IP is no longer free. The foundational 2023 patent reads on every public-Ethereum tokenization platform, every L2 rollup, every cross-chain messaging operator. The 2026 PQ family extends coverage through the regulatory PQ migration window. Lux Industries' enforcement program against non-members begins in 2026.

First-mover field-of-use positioning. Strategic and Founding membership tiers include most-favored-licensee protection within the member's selected field. Early members lock in pricing and field-of-use exclusivity that later entrants cannot replicate.

9 Contact

- W3A Foundation membership inquiries — `members@w3a.foundation`
- Lux Industries CEO, Zach Kelling — `z@lux.industries`
- Lux Industries general counsel — `counsel@lux.industries`
- Institutional partnerships — `partners@lux.industries`
- Substrate engineering — `platform@lux.industries`

We respond to membership inquiries within five business days. Initial executive conversations scheduled within fourteen days of inbound.

Confidentiality and reservation of rights

Lux Industries reserves all rights with respect to the patent estate, including the right to assert any patent against any party operating outside an executed W3A Foundation membership and license. Nothing herein constitutes a covenant not to sue, a license grant, or a waiver of any patent rights. The patent license described herein is available exclusively through executed W3A Foundation membership.

W3A Foundation — `w3a.foundation`
Lux Industries, Inc. — Delaware
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